

1.

```
if(pid == 0)
{
    printf("Child PID : %d\n", getpid());
    execlp(cmd, cmd, NULL);
    perror("Execution Failed");
}
else if(pid > 0)
{
    printf("Parent PID : %d\n", getpid());
    wait(NULL);
    printf("Child process completed.\n");
}
else
{
    printf("Fork Failed\n");
}

return 0;
}
```

```
pooh@Kandulas-MacBook-Air ~ % nano childprocess.c
pooh@Kandulas-MacBook-Air ~ % gcc childprocess.c -o childprocess
pooh@Kandulas-MacBook-Air ~ % ./childprocess
Enter Linux command: ls
Parent PID : 50777
Child PID : 50782
a.out
blenderkit_data
childprocess
childprocess.c
class Nodef.java
CO-1.pdf
CO-2.pdf
CO-3.pdf
Completion certificate react.pages
Completion certificate react.pdf
Desktop
Documents
Downloads
example
example.c
fedf pdf 2.pdf
FEDF_lab_Programs insem .pdf
FEDF_lab_Programs pdf.pdf
FEDF_lab_Programs.pages
first.blend
fork
fork_exec
fork_exec.c
fork.c
hello
hello.c
import java.util.java
Library
Movies
Music
OS
parent_child
parent_child.c
pdf fedf.pdf
PharmaInventory.java
Pictures
PID
PID.c
Poojitha Resume.pages
process
process.c
Project abstract submission form-FEDF (2).pages
Project abstract submission form-FEDF (2).pdf
Public
PyCharmMiscProject
PycharmProjects
read_user_name
read_user_name.c
RetailMind - Smart Retail Inventory & Sales Analytics System
Untitled.ipynb
```

```
Project abstract submission form-FEDF (2).pdf
Public
PyCharmMiscProject
PycharmProjects
read_user_name
read_user_name.c
RetailMind - Smart Retail Inventory & Sales Analytics System
Untitled.ipynb
VF.html
wait
wait.c
Child process completed.
[pooh@Kandulas-MacBook-Air ~ % ./childprocess
Enter Linux command: lscpu
Parent PID : 50789
Child PID : 50791
Execution Failed: No such file or directory
Child process completed.
pooh@Kandulas-MacBook-Air ~ % █
```

```
[pooh@Kandulas-MacBook-Air ~ % ./childprocess
Enter Linux command: lsblk
Parent PID : 50796
Child PID : 50798
Execution Failed: No such file or directory
Child process completed.
pooh@Kandulas-MacBook-Air ~ % █
```

```
[pooh@Kandulas-MacBook-Air ~ % ./childprocess
Enter Linux command: ps
Parent PID : 50802
Child PID : 50803
  PID TTY          TIME CMD
 50542 ttys000    0:00.12 -zsh
 50802 ttys000    0:00.01 ./childprocess
Child process completed.
pooh@Kandulas-MacBook-Air ~ % █
```

Terminal Shell Edit View Window Help										Tue Aug 11 11:05 AM												
Processes: 496 total, 4 running, 492 sleeping, 2684 threads																				11:05:10		
Load Avg: 2.49, 3.42, 3.78 CPU usage: 10.52% user, 8.30% sys, 81.16% idle SharedLibs: 428M resident, 90M data, 60M linkedit, MemRegions: 0 total, 0B resident, 0B private, 631M shared.																						
PhysMem: 7434M used (1479M wired, 2530M compressor), 190M unused. VM: 213T vszize, 6144M framework vszize, 2791359(8) swaptins, 3883782(0) swaptouts. Networks: packets: 39626507/150 in, 4226052/1227M out.																						
Disks: 27958608/659G read, 954700/175G written.																						
PID	COMMAND	%CPU	TIME	#TH	#WO	#PORTS	MEM	PURG	CMRPS	PGRP	PPID	STATE	BOOSTS	%CPU_ME	%CPU_OTHRS	UID	FAULTS	COW	MSGSENT	MSGRCV	SYSBSD	SYSMACH
403	WindowServer	46.8	06:56:38	21	6	5221-	711M+	39M-	284M+	403	1	sleeping	*0[1]	0.57275	2.08526	88	25431189+	91147	219846462+	246311993+	421023825+	742651798+
0	kernel task	35.6	04:31:01	579/9	0	0	412M+	0B	0B	0	0	running	0[0]	0.00000	0.00000	0	23960	10	269704057+	199838481+	0	0
634	talagentd	24.0	01:02:00	2	1	202	11M+	0B	5760K	634	1	sleeping	*0[4006]	0.00000	24.03695	501	789172+	190	136496+	57656+	167640+	299097+
49540	Telegram	8.1	06:23:02	33	5	891+	279M	32K	238M	49540	1	sleeping	*0[1348+]	0.69916	0.01447	501	998787+	1100	4680049+	463624+	3685922+	8697281+
50809	top	6.0	06:01:67	1/1	0	29-	7953K	0B	0B	50808	50808	running	*0[1]	0.00000	0.00000	0	7245+	70	668527+	334276+	40931+	388687+
474	mONSResponse	4.8	26:37:15	6	4	130	11M+	0B	2192K	474	1	sleeping	*0[1]	0.00000	0.00000	65	729963+	159	2127748+	1006568+	50546445+	3312930+
49975	Google Chrom	4.3	01:03:41	16/1	1	132	39M	0B	20M	49967	49967	running	*2186[173]	0.00000	0.00000	501	49334	856	73859	30587	7841050+	127663
49513	Terminal	3.6	02:06:53	11	4	465+	154M+	41M+	13M	49513	1	sleeping	*0[757+]	24.5168	0.05063	501	112776+	405	280705+	58009+	217094+	731919+
46818	WhatsApp	2.4	08:18:01	30	6	711	175M	640K	138M	46818	1	sleeping	*0[15824]	0.00000	0.00000	501	385259	3360	1074560+	516811+	4751308+	173053000+
513	com.apple.Dr	2.2	40:09:16	7	5	1411	29M	0B	4928K+	513	1	sleeping	0[1]	0.00000	0.00000	270	7887166	63	24012072+	13266327+	44649803+	51534891+
728	Finder	2.0	12:23:45	8	5	1735	334M	288K	286M+	728	1	sleeping	*0[18135+]	0.00000	1.94012	501	4801113+	568	4739173	2181955+	5486967+	9979361+
655	sharingd	1.9	05:57:92	4	1	415	20M+	0B	11M	655	1	sleeping	*0[1]	1.94012	0.00000	501	865104+	321	995251+	869337+	3891199+	5387577+
601	WindowManage	1.2	04:19:08	6/1	3	456-	22M+	0B	8912K-	601	1	running	*0[55169+]	0.00000	0.00001	501	864122+	202	925410+	342942+	1409575+	3015587+
46900	Safari	0.7	07:59:40	14	4	1001	154M	0B	124M	46900	1	sleeping	*0[17708+]	0.00338	0.01307	501	1055269+	5444	9055216+	2415342+	5730918+	19560445+
50813	screencaptur	0.6	00:00:11	6	4	192-	8913K+	0B	0B	50813	1	sleeping	*0[69+]	0.00698	0.17931	501	2167+	185+	1078+	353+	2926+	2710+
50673	netbiosd	0.5	00:01:47	7	6	40	3009K	0B	592K	50673	1	sleeping	*0[1]	0.00000	0.00000	222	1785	110	236	70	76913+	830+
526	mysqld	0.4	07:28:11	38	0	63	365M	0B	363M	526	1	sleeping	*0[1]	0.00000	0.00000	74	68851	130	299874+	149910+	35736568+	317029+
330	logd	0.3	12:45:78	4	3	1645-	23M+	0B	24M+	330	1	sleeping	*0[1]	0.00000	0.00000	0	1767269+	92	4327496+	4872566+	23712471+	4883384+
334	fsevents	0.2	07:16:23	13	1	191	5233K	0B	2176K+	334	1	sleeping	*0[1]	0.00000	0.00000	0	1300980+	88	2173651+	625919+	18289899+	3527903+
393	corebrightness	0.2	12:24:71	3	2	145	464K	0B	2000K	393	1	sleeping	*0[1]	0.13959	0.00000	0	583644+	130	2662508+	2082413+	23061470+	5021190+
584	distnoted	0.2	02:36:44	3	2	555-	4625K	0B	1632K+	584	1	sleeping	*0[1]	0.00000	0.19990	501	177859	87	619585	1286285+	8036454+	1516677+
4584	Spotify	0.1	17:07:68	59	3	886	209M	0B	148M-	4584	1	sleeping	*2[46958+]	0.00000	0.00000	501	6876757+	1308	2300547+	1068833+	39702097+	6143568+
50721	Google Chrom	0.1	00:24:56	21	1	258	293M	0B	201M	49967	49967	sleeping	*2[21953]	0.00000	0.00000	501	101443	8379	102539	60632	139061	215537+
4592	Spotify Help	0.1	01:40:31	17	1	116	30M+	0B	20M	4584	4584	sleeping	*2[21995+]	0.00000	0.00000	501	653526+	872	623845+	313885+	2163960+	1177097+
47782	SiriNServic	0.1	00:09:00	7	3	277	7329K	0B	3616K-	47782	1	sleeping	*0[15173+]	0.00000	0.14230	501	42716+	191	143350+	53759+	268567+	371106+
385	distnoted	0.1	00:45:28	3	2	176	2337K	0B	1056K	385	1	sleeping	*0[1]	0.00000	0.12542	241	109000+	87	908174+	131810+	1010999+	1043043+
484	audioclocksy	0.1	02:42:67	4	3	53	3073K	0B	1552K	484	1	sleeping	*0[1]	0.05315	0.00000	0	31287	89	1844644+	1530301+	3601499+	3181078+
4700	Siri	0.1	00:27:00	7	3	297	10M	0B	5584K	47700	1	sleeping	*0[6758+]	0.00000	0.12275	501	80065	205	199195+	60299+	367784+	522888+
344	powerd	0.1	03:58:49	3	2	164	5649K+	0B	1376K+	344	1	sleeping	*0[1]	0.00000	0.07115	0	1167574+	122	3219383+	1990104+	6228579+	5552055+
456	airportd	0.1	02:33:44	8	6	391-	21M+	0B	4768K+	456	1	sleeping	*1[1]	0.00000	0.00000	0	10327641	285	43793102	22802223+	140230009+	47333865+
2872	bluetoothd	0.1	09:15:94	11	5	452	14M	256K	6368K	2872	1	sleeping	*0[1]	0.47064	0.03045	0	573179+	190	1284122+	987001+	17557599+	2749393+
421	coreaudiod	0.0	40:44:50	15	8	234M-	20M+	0B	11M	421	1	sleeping	*0[1]	0.00000	0.03239	202	7087466+	238	53301701	49613192+	21045203+	152075121+
373	launchservi	0.0	07:42:54	5	4	767-	7633K+	0B	1344K	373	1	sleeping	*3[10855249]	0.00000	0.04257	0	1103389	103	5474192+	4330087+	15741712+	9392512+
739	ViewBridgeAu	0.0	00:38:17	3	2	402	4577K	0B	2496K	739	1	sleeping	*0[141418+]	0.00000	0.03685	501	307155+	113	484067+	206204+	924514+	1286002+
405	loginwindow	0.0	34:49:42	7	4	842	65M	0B	48M	405	1	sleeping	*436[12706]	0.00250	0.01494	501	5263172+	4524	19283054+	5131607+	18219395+	48162670+
5803	duetexpertd	0.0	22:56:45	3	2	446	22M	0B	20M	5803	1	sleeping	*0[20222]	0.00000	0.00000	501	1757252	408	300593+	482639+	29547179+	2443242+
729	fontd	0.0	01:19:79	3	2	166-	13M	0B	10M	729	1	sleeping	*0[56542]	0.00000	0.00000	501	480727	105	102090	102052+	1125892+	279573+
46202	PerfPowerSer	0.0	02:52:00	7	4	776	14M	256K	8112K	46202	1	sleeping	*0[13038]	0.00000	0.00000	0	127224	170	1428263	940800+	2284944+	2209636
396	notified	0.0	03:05:25	2	1	902-	4321K+	0B	720K	396	1	sleeping	*0[1]	0.00000	0.00000	0	158541	60	2261691	2980600+	10773555+	5357280+
46835	Pages	0.0	02:43:20	8	3	856	414M	0B	327M	46835	1	sleeping	*0[1926]	0.00000	0.00000	501	460526	850	1215407	297276	5300074+	2907723
378	locationd	0.0	18:50:55	5	2	306-	11M+	0B	5776K-	378	1	sleeping	*0[42452]	0.00000	0.00000	205	926493+	184	9812850+	4999183+	8741001+	7085863+
1	launchd	0.0	17:54:45	5	3	434-	25M	0B	6464K	1	0	sleeping	*0[0]	0.00000	0.00000	0	1640301	70	3309279	3229823+	17537532+	9061755+
46447	photoanalsi	0.0	00:04:04	3	2	163+	16M+	0B	15M	46447	1	sleeping	*0[1]	0.00000	0.00000	501	28359+	259	7238	1322	79209+	32829
453	distnoted	0.0	00:06:00	3	2	30	1681K	0B	976K	453	1	sleeping	*0[1]	0.00000	0.00000	88	67048+	87	225	44576+	521067+	44890+
687	fileprovider	0.0	02:31:45	5	4	172	10M	0B	10M	687	1	sleeping	*0[91]	0.00000	0.00000	501	353732	221	176359	163560	5200976+	612856
570	distnoted	0.0	00:06:04	3	2	30	1681K	0B	976K	570	1	sleeping	*0[1]	0.00000	0.00730	242	67196+	87	223	44557+	520499+	44870+
501	distnoted	0.0	00:06:25	3	2	30	1665K	0B	960K	501	1	sleeping	*0[1]	0.00000	0.00847	262	66410+	87	223	44577+	509110+	44897+
723	Dock	0.0	02:12:07	5	3	893	63M	0B	55M+	723	1	sleeping	*1[4752]	0.00000	0.01821	501	901700+	269	1893309+	328123	1692304+	4242611+
457	distnoted	0.0	00:06:61	3	2	30	1713K	0B	1008K	457	1	sleeping	*0[1]	0.00000	0.00938	202	67621+	87	216	44576+	520964+	44894+
467	distnoted	0.0	00:06:72	3	2	30	1681K	0B	992K	467	1	sleeping	*0[1]	0.00000	0.00882	0	70597+	87	301	44717+	512246+	45200+

2.

```

pooh@Kandulas-MacBook-Air ~ % nano child.c
pooh@Kandulas-MacBook-Air ~ % gcc child.c -o child
pooh@Kandulas-MacBook-Air ~ % nano input.txt
pooh@Kandulas-MacBook-Air ~ % ./child
Enter Linux command: ls
Parent PID : 51102
Child PID : 51108
a.out
blenderkit_data
child
child.c
childprocess
childprocess.c
class NodeDef.java

```

